

TuniCars+

Resilience Testing Report

Docker Swarm vs Kubernetes

Session ID

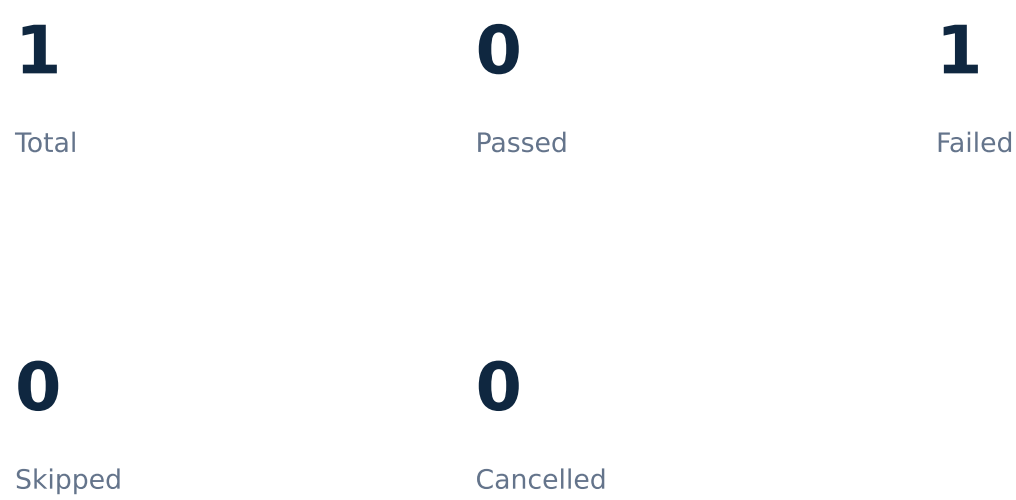
20260813_152822_301286

Date

2026-08-13T14:28:22.137716+00:00

Executive Summary

20260813_152822_301286



Overall measured availability: 100.00%

During this campaign, 0 of 1 scenarios passed, 1 failed and 0 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.

Test Environment

Architecture and technologies

Frontend → Backend → PostgreSQL

Docker Swarm environment

- Docker services
- FastAPI backend
- Next.js frontend
- PostgreSQL database

Kubernetes / Minikube environment

- Kubernetes Deployments and Services
- Minikube nodes
- FastAPI · Next.js · PostgreSQL

Test Scenarios

Scenario	swarm
cpu	1

Full Results Table

Scenario	Platform	Detection	Recovery	Required	Success	Avg ms	P95 ms	HTTP Fail	Avail. %	Status	Skip Reason	Error Message
cpu	swarm	N/A	N/A	N/A	N/A	12.077	18.814	N/A	100.0	FAIL	N/A	Swarm stress container exited with code 1.

swarm — cpu

State: Fail
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Swarm stress container exited with code 1.

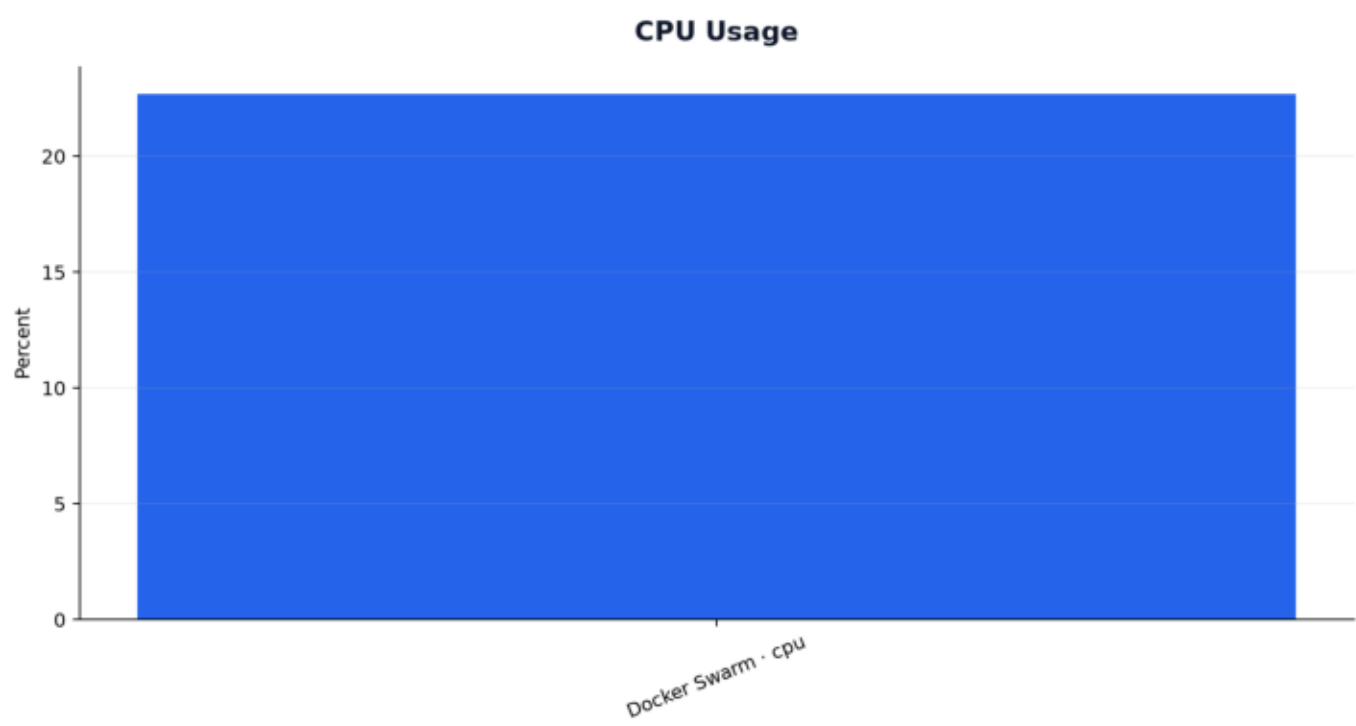
Availability

Current-session measured data



Cpu Usage

Current-session measured data



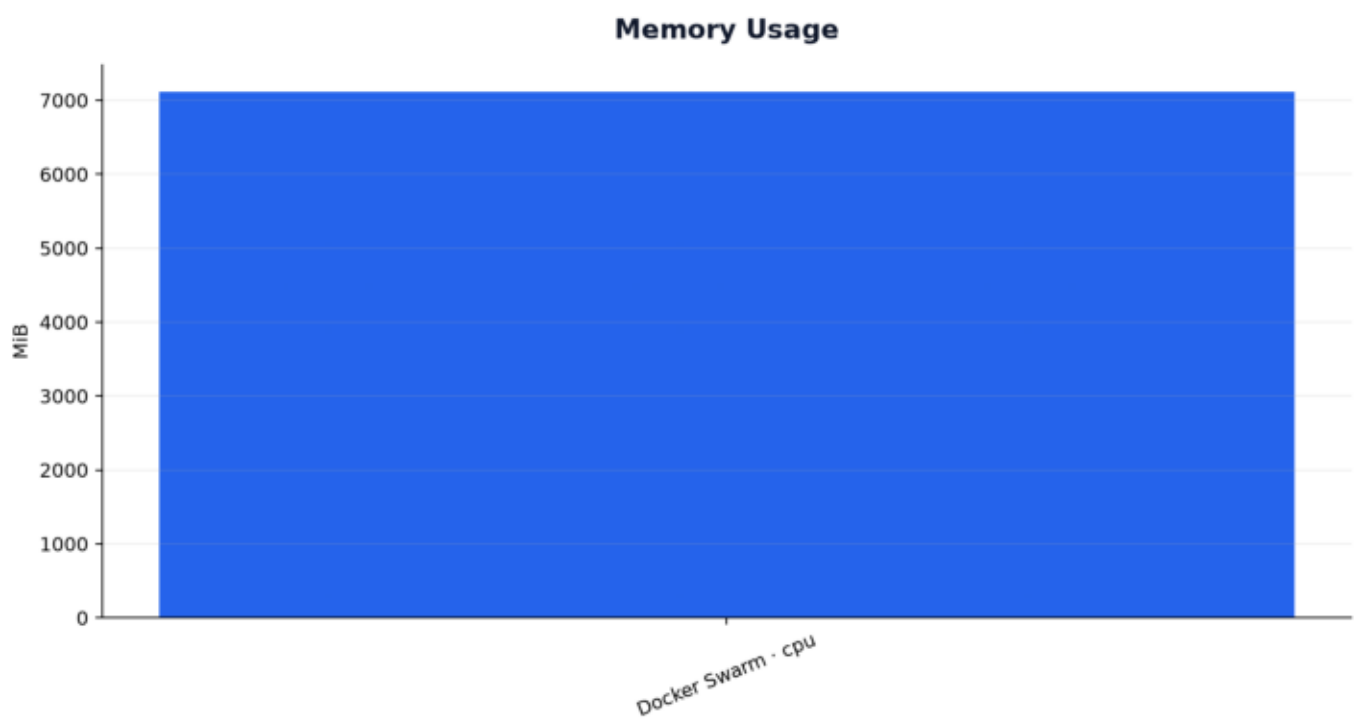
Http Failures

Current-session measured data



Memory Usage

Current-session measured data



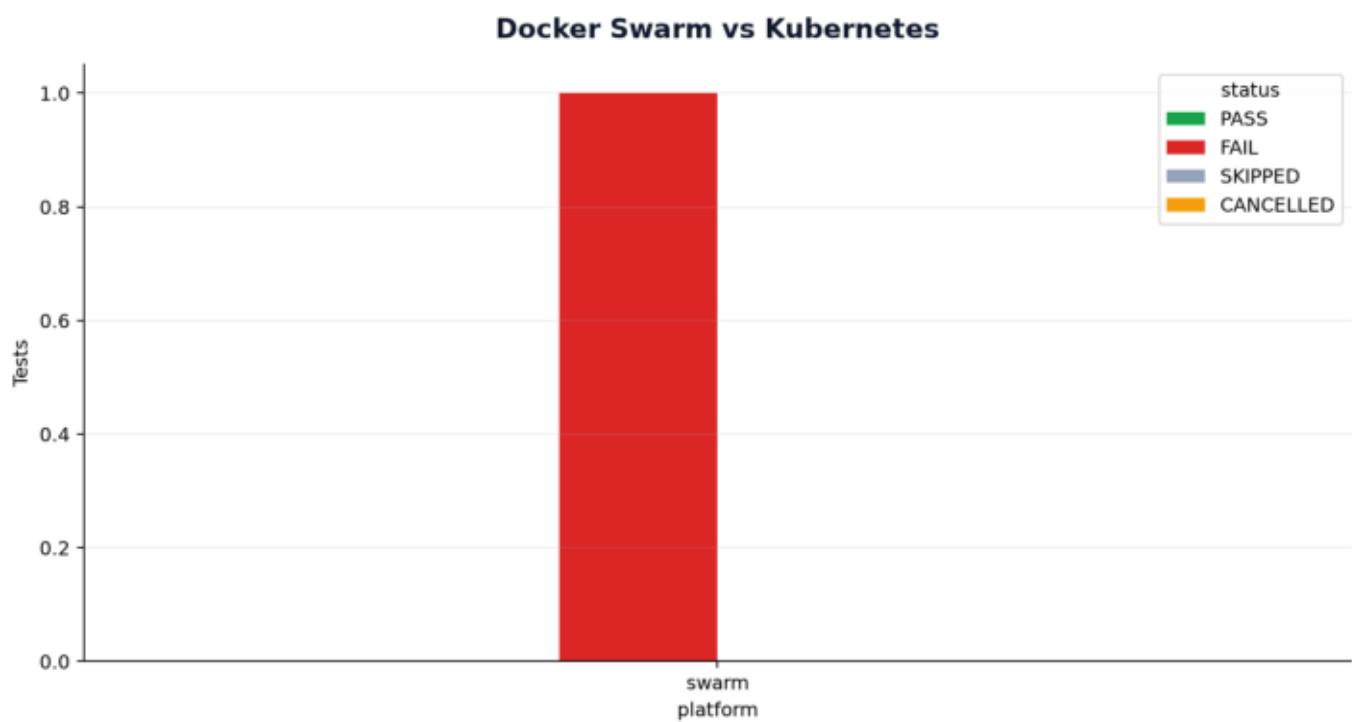
P95 Response Time

Current-session measured data



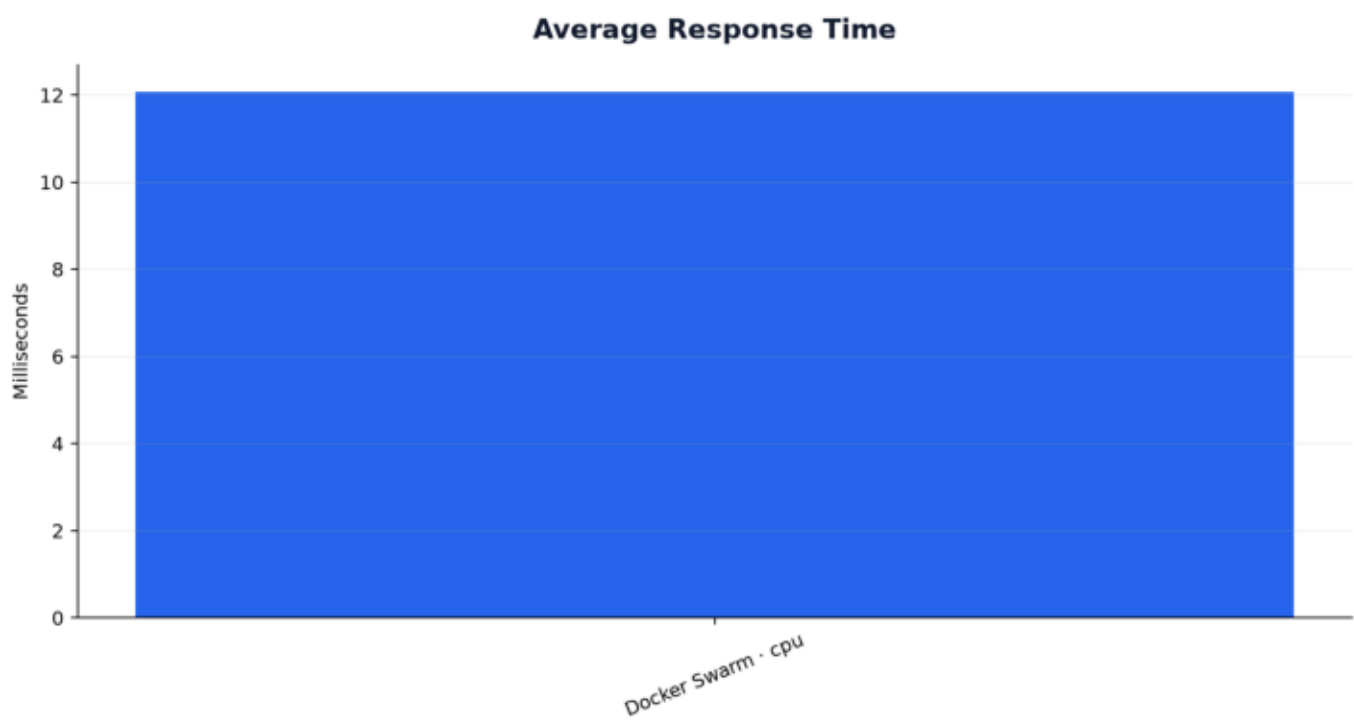
Platform Comparison

Current-session measured data



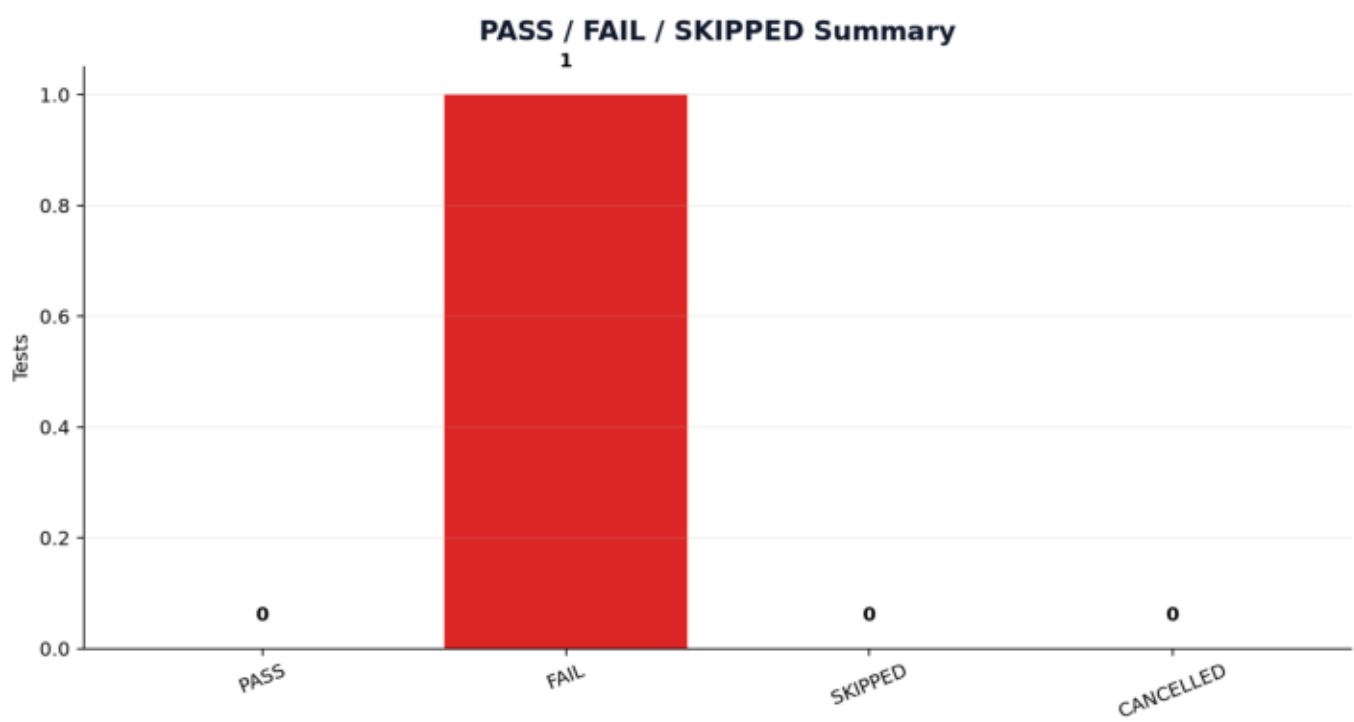
Response Time

Current-session measured data



Status Summary

Current-session measured data



Failures and Errors

Platform	Scenario	Status	Error
swarm	cpu	FAIL	Swarm stress container exited with

Skipped Tests

Platform	Scenario	Exact reason
—	—	No skipped tests recorded.

Swarm vs Kubernetes Comparison

Platform	FAIL
swarm	1

Final Conclusion

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During this campaign, 0 of 1 scenarios passed, 1 failed and 0 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.